Chinmoy Mondol

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Google Scholar /chinmoymondol /chinmoy-mondol

Research Interests

My research centers on **computer vision**, **multimodal AI**, and **agentic AI**, with a growing interest in efficient perception for real-world systems. Recent work spans vision-transformer optimization for edge deployment, multimodal context understanding, and RAG-based conversational evaluation.

Education

BSc. in Computer Science and Engineering Sept 2017 – April 2021
American International University-Bangladesh (AIUB) *Dhaka, Bangladesh*

- CGPA: 3.99/4.00
- Thesis: "Music Suggestions from Determining the Atmosphere of Images" (Supervisor: Md Kishor Morol)

Research Experience

Research Assistant - Remote Sept 2026 – Present
ELITE Research Lab | Supervisor: Abir Bokhtiar *Queens, NY, USA*

- Build an open Landsat/Sentinel benchmark (1988–2026) for forecasting riverbank erosion on Bangladesh's Jamuna and Padma rivers, with three forecasting tasks and baselines from persistence to a geospatial foundation model.
- Develop IndicTextBench, a benchmark for visual text rendering in Bengali, Devanagari, and Tamil, evaluating text-to-image diffusion models with OCR, grapheme-cluster error rates, and a native-reader validation study.
- Contribute to a vulnerability assessment of medical multimodal LLMs, extending an adversarial text-image evaluation framework toward 2,500+ clinical cases.

Publications

- **Mondol, C.**, et al., "Real-Time Human Pose Estimation for Collaborative Robotics Using ROS 2 and Edge-Optimized Transformers," submitted to ICEEICT 2027.
- "A Multi-Decade Satellite Benchmark for Forecasting Riverbank Erosion in Large Braided Rivers," in preparation, 2026.
- "Shaping Matters: Benchmarking and Fixing Visual Text Rendering for Indic Scripts in Diffusion Models," in preparation, 2026.
- Nuhel, A. K., Taha, A. S., Ibrahim, M., Oishy, M. I., **Mondol, C.**, et al., "Integrating Visual Data in Conversational AI: A Study on LangChain Custom Tools for Image-Based Interaction," 2025 IEEE International Conference on Emerging Technologies and Applications (MPSec ICETA), Gwalior, India, 2025, pp. 1–6, doi: 10.1109/MPSecICETA64837.2025.11118796.
- Sohel, S. I., **Mondol, C.**, Ayon, H. S., Islam, U. T., & Morol, M. K., "Music Suggestions from Determining the Atmosphere of Images," 2021 24th International Conference on Computer and Information Technology (ICCIT), Dhaka, Bangladesh, 2021, pp. 1-7, doi: 10.1109/ICCIT54785.2021.9689781.

Research & Technical Skills

- **Research Skills:** Data Analysis, Literature Review, Academic Writing
- **Programming Languages:** Python, C, C++, Java
- **Machine Learning:** TensorFlow, Keras, scikit-learn, OpenCV, NumPy, Pandas, Jupyter Notebook, Kaggle
- **Databases:** SQL (MySQL, PostgreSQL), NoSQL (Redis)
- **Systems & Infrastructure:** Docker, Git
- **Writing Tools:** LaTeX

Awards & Honors

- Received a **Gold Medal** at the convocation ceremony in recognition of outstanding academic performance. 2021
- Graduated with the highest academic distinction, **Summa Cum Laude**. 2021
- Awarded **Dean's List Honors** four times during my tenure at AIUB. 2017 – 2021
- Awarded **Merit-Based Academic Scholarship (Tuition Fee Waiver)** from AIUB. 2017 – 2021
- Won **1st prize** in Inter-University ICT Olympiad at AIUB CS Fest. 2018

Teaching Experience

Teaching Assistant Fall 2021

American International University-Bangladesh (AIUB) *Dhaka, Bangladesh*

- Served as a TA in the "Artificial Intelligence and Expert System (CSC4121)" and "Web Technologies (CSC3222)" courses, assisting 12 lab sessions each and supporting 70+ undergraduate students across both.
- Designed programming assignments covering search strategies, adversarial search, and genetic algorithms (AI course), and tokenization and MVC architecture (Web Technologies)
- Assisted faculty members with grading assignments and final projects.

Professional Experience

Assistant Lead Engineer (QA) Sept 2025 – Present

bKash Limited *Dhaka, Bangladesh*

- Evaluate RAG-based conversational systems for a fintech assistant, building multi-dialect (Bangla/Banglish/English) test suites to stress-test retrieval and dialogue accuracy, and validating ASR/TTS pipelines for accent and financial-term precision.

Software Engineer (QA) → Senior Software Engineer (QA) Jul 2021 – Sept 2025

Tiger IT Bangladesh Limited *Dhaka, Bangladesh*

- Validated a real-time vehicle number plate recognition system, focusing on inter-service communication and Deep SORT integration.
- Collaborated with DevOps teams on containerized deployments, gaining hands-on exposure to Docker-based environments and deployment workflows.

References

Md Kishor Morol
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